

Letter of Recommendation for Fourie Van Rooyen

To Whom It May Concern,

I am pleased to write this letter of recommendation for Fourie Van Rooyen, an outstanding candidate for any engineering position. My name is Matthew Cotting, and as an aerospace engineering student and the Lead of the Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.) at The University of Alabama, I have worked closely with Fourie for over two years. As my dedicated team member and engineering colleague, Fourie has consistently demonstrated exceptional technical skills, leadership, and a passion for aerospace engineering.

Fourie possesses a robust skill set that includes proficiency in SolidWorks, ANSYS, MATLAB, and Python, alongside expertise in aerodynamics, propulsion systems, and finite element analysis (FEA). One of his standout contributions was leading the design of a propeller with vortex inductors using SolidWorks as part of the A.A.E.R.O. team. He conducted advanced simulations in ANSYS to assess thrust and acoustic performance, applying FEA and GD&T principles to ensure structural integrity. These designs were rigorously validated through wind tunnel testing and static test stand experiments, showcasing his ability to translate concepts into practical outcomes.

In addition to his technical prowess, Fourie excels in project management and team collaboration. As the Lead Simulation and Testing Engineer for A.A.E.R.O., he oversaw the experimental validation of propulsion systems, meeting project deadlines ahead of schedule. His leadership extended to the Alabama Rocketry Association, where he contributed to the design and testing of a single-stage liquid propellant rocket. Fourie's hands-on experience with hydro-testing, machining graphite nozzles, and hot fire testing of a LOX and Kerosene engine highlights his practical engineering capabilities and meticulous attention to detail.

Fourie's soft skills further enhance his qualifications. His problem-solving abilities, effective communication, and time management have been invaluable in fostering a productive team environment. Whether troubleshooting complex issues or mentoring peers, he approaches every task with professionalism and a solution-oriented mindset.

Beyond academic projects, Fourie has gained practical experience through roles at Trinity Farm and National Tire and Battery (NTB). These positions strengthened his mechanical aptitude, customer service skills, and organizational abilities, making him a versatile candidate for engineering roles.

Fourie's dedication to the field is evident in his pursuit of certifications, including the Certificate in Aerospace Engineering: Aircraft Fundamentals and the Certificate in Rocket

Engineering and Interstellar Space Propulsion. These accomplishments, paired with his hands-on experience, position him as an ideal fit for roles in aerospace design, propulsion systems, aerodynamics analysis, or engineering project management.

In conclusion, Fourie Van Rooyen is a highly skilled and motivated engineer who will undoubtedly excel in any professional setting. I wholeheartedly endorse him for your consideration and am confident he will make a valuable contribution to your organization.

Sincerely,

Matthew Cotting

Aerospace Engineering Student & A.A.E.R.O. Team Founder

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